

ACA 2003: Preliminary Schedule of Sessions and Talks

(Last Updated July 18, 2003)

Session Number	Session Name	Organizers	# of Talks in Session
T1	Computational aspects of algebraic curves	Tony Shaska (UC Irvine)	11
T2	Symbolic Summation	Sergei A. Abramov (Moscow), Marko Petkovsek (Ljubljana), Eugene V. Zima (Waterloo)	9
T3	Computer Algebra in Education	Alkis Akritas (Thessaly, Greece), Michael Wester (New Mexico), Bill Pletsch (New Mexico)	7
T4	Computer Algebra in Science and Technology	Laurent Bernardin (Maplesoft), David Jeffrey (Western Ontario)	4
T5	Groebner Bases and Applications	Quoc-Nam Tran (Lamar University, USA) and Alexander Levin (Catholic University of America)	10
T6	Interval Computation	Ned Nedialkov (McMaster) and George Corliss (Marquette)	8
T8	Elimination Theory	Amit Khetan and Carlos D'Andrea (UC Berkeley, USA)	9
T9	Computer Algebra and Polynomial Systems in Chemistry	Karin Gatermann (Berlin)	5
T10	Symbolic Linear Algebra	B. David Saunders (Delaware) and Gilles Villard (ENS Lyon)	8
T11	Mathematics on the Internet	Mike Dewar (NAG, UK)	8
T12	Symbolic-Numeric methods for Curves and Surfaces	J. Schicho and Mohamed Shalaby (RISC Linz, Austria)	9
T13	Young Investigators	Mark Giesbrecht (Waterloo) and Stanly Steinberg (New Mexico)	7

Conference Overall Schedule

Date	<u>July 28, 2003</u> <u>Monday</u>			<u>July 29, 2003</u> <u>Tuesday</u>			<u>July 30, 2003</u> <u>Wednesday</u>			July 31, 2003 Thursday
Start Time	A	B	C	A	B	C	A	B	C	
9:00	Official Opening			T5	T6	T8	RedHat Tour 			Panel Discussion
9:30	T1	T2	T3	T5	T6	T8				
10:00	T1	T2	T3	T5	T6	T8				
10:30	Break			Break						Trip to Wrightsville Beach 
11:00	T1	T2	T3	T5	T6	T8	T12	T10	T11	
11:30	T1	T2	T3	T5	T6	T8	T12	T10	T11	
12:00	T1	T2	T3	T5	T6	T8	T12	T10	T11	
12:30	Lunch			Lunch			Lunch			
1:00										
1:30										
2:00	T1	T2	T3	T9	T6	T8	T12	T10	T11	
2:30	T1	T2	T3	T9	T6	T8	T12	T10	T11	
3:00	T1	T2		T9		T8	T12	T10	T11	
3:30	Break			Break			Break			
4:00	T1	T13	T4	T9	T5	T13	T12	T10	T11	
4:30	T1	T13	T4	T9	T5	T13	T12	T10	T11	
5:00	T1	T13	T4	T9	T5	T13	T12	T10		
			T4		T5	T13				
	Activity			Activity (Banquet)			Activity			End of Conference

Monday July 28, 2003

Start Time	A	B	C
9:00	Official Opening		
9:30	B. Brock: <i>Moments associated to the moduli space of hyperelliptic curves over a finite field</i>	Juergen Gerhard, <i>Ha Le Symbolic summation in Maple</i>	Jerry Uhl, <i>Raising the Level of Precalculus Mathematics</i>
10:00	Burhanuddin, <i>On computing discrete logarithms in Formal groups and it's applications</i>	Burkhard Zimmermann, <i>Summation in Difference Differential Rings</i>	Eugenio Roanes-Lozano & Eugenio Roanes-Macias, <i>Symbolic Manipulation in Projective Geometry through the Cooperation of Dynamic Geometry Systems and Computer Algebra Systems</i>
10:30	Break		
11:00	J. Wolper, <i>Theta Vanishings and Automorphism Groups of Riemann Surfaces of Genus Three</i>	Sergei A. Abramov, Marko Petkovsek <i>Gosper's and Zeilberger's algorithms revisited</i>	Laurent Bernardin, <i>A Look at the New Student Packages in Maple</i>
11:30	T. Shaska, <i>Computational Aspects of Algebraic Curves, a survey</i>	Mark Giesbrecht, <i>Fast p-adic algorithms for computing dispersion</i>	Michel Beaudin, <i>Give Mathematics a Chance!</i>
12:00	C. Yin, <i>Special Loci in Moduli of Curves</i>	Carsten Schneider, <i>Symbolic Summation in # - #Fields</i>	Karsten Schmidt & Wolfgang Moldenhauer, <i>Using the TI-89 in Math Education - What are the Students' Views?</i>
12:30	Lunch		
2:00	M. Seppala, <i>Numerical uniformization of algebraic curves</i>	Victor Adamchik, <i>Symbolic Computation of Series and Products in Terms of the Multiple Gamma and Related Functions</i>	Bill Pletsch, <i>A CAS Lecture for the Calculus Students: The Riemann Product</i>
2:30	D. Mills, <i>Some Questions Regarding Avoidable Families of Sets</i>	Mark van Hoeij, <i>Rational definite summation</i>	Giovanna Albano & Matteo Desiderio, <i>An Effective Learning using CAS</i>
3:00	G. Cardona, <i>Curves of genus 2 over arbitrary fields</i>	Arne Storjohann & Eugene Zima, <i>Rational function reconstruction and closed form summation</i>	
3:30	Break		
4:00	C. Hurlburt, <i>Computational Aspects of Differential Algebraic Geometry</i>	Young Investigator: Howard Cheng, <i>Algorithms for Normal Forms for Matrices of Ore Polynomials</i>	Paulina Chin, <i>Automatic Code Generation</i>
4:30	M. van Hoeij, <i>Reparametrizing rational algebraic curves</i>	Young Investigator: Yang Zhang, <i>Popov forms of Ore matrices</i>	Cecelia Laurie, <i>Statistical Genetics</i>
5:00	T. Yamauchi, <i>On Q-simple factors of $J_0(N)$</i>	Young Investigator: Alin Bostan, <i>Tellegen's principle into practice.</i>	Brian Moore, <i>Symofros: A symbolic modeling and simulation tool for mechanical system</i>
5:30			Joseph Schicho, <i>Improving the element pre-conditioning method by symbolic computation</i>
	Activity		

Tuesday July 29, 2003

Start Time	A	B	C
9:00	Yosuke Sato, <i>ACGB on Varieties</i>	Ned Nedialkov, <i>Introduction to Interval Numerical Methods</i>	Agnes Szanto, <i>Elimination theory of non-commutative algebras of differential invariants</i>
9:30	Alexander Levin, <i>Groebner Bases w.r.t. Several Orderings and Difference Dimension Polynomials</i>	John Pryce, <i>Mapping Expressions to Functions in Cset Interval Arithmetic</i>	John Nahay, <i>Differential Resolvents of Minimal Order and Weight</i>
10:00	V. Mityunin, Pankratev, <i>Comparison of the parallelization quality of algorithms for computing Groebner and involutive bases</i>	Nathalie Revol, <i>Multiple Precision Interval Arithmetic and Application to Linear Systems</i>	Arthur Chtcherba, <i>Sylvester-type matrices via the Dixon resultant formulation and their optimizations</i>
10:30	Break		
11:00	Quoc-Nam Tran, <i>Efficient Groebner Basis Computation for Finding Implicit Representations of Geometric Objects</i>	Markus Neher, <i>From Intervals to Taylor Models: A Numeric Symbolic Approach to Validated Computation</i>	Manfred Minimair, <i>Developments in resultants of composed polynomials</i>
11:30	Martnez-Moro, <i>Using Groebner basis for determining the equivalence of linear codes.</i>	Jeff Tupper, <i>Pictures from Proofs: Sound Graphing Algorithms</i>	Amit Khetan, <i>Implicitization of rational surfaces using toric varieties</i>
12:00	Ekaterina Shemyakova, <i>Graphs of involutive divisions</i>	Ekaterina Auer, <i>Interval Arithmetic in the Multibody Modeling System MOBILE</i>	Laurent Busé, <i>Implicitizing rational hypersurfaces using approximation complexes</i>
12:30	Lunch		
2:00	Anke Sensse, <i>Electrocatalytic oxidation of formic acid</i>	Yves Papegay, <i>ALIAS: A Library Mixing Interval Analysis and Computer Algebra</i>	Marc Moreno Maza, <i>Recent advances in triangular decomposition methods</i>
2:30	Brandilyn Stigler, <i>Polynomial Models for Gene Regulatory Networks</i>	Stan Wagon, <i>The Role of Symbolic Computation in the SIAM/Oxford 100-Digit Challenge</i>	Ilias Kotsireas, <i>Recent advances in polynomial system solving and an application in Chaos Theory</i>
3:00	Karin Gatermann, <i>Positive solutions of systems with mass action kinetics</i>		Ming Zhang, <i>An Approximation Approach to Molecular Conformational Search</i>
3:30	Break		
4:00	Carsten Conradi, <i>Model Discrimination using Algebraic Geometry and Computer Algebra Systems</i>	Farr, Gao, <i>Computing Groebner Bases for Vanishing Ideals of Finite Sets of Points</i>	Young Investigator: Alexey Ovchinnikov, Moscow State University, <i>Characterizable Radical Differential Ideals and Characteristic Sets</i>
4:30	David M. Malonza, <i>Groebner basis methods in Symbolic Computation of Invariants and Equivariants applied to Normal forms of Non linear Systems with Nilpotent linear part</i>	Aleksandra Slavkovic: <i>An Application of Algebraic Geometry in Statistical Disclosure Limitation</i>	Young Investigator: Hirokazu Anai, <i>On Solving Real Algebraic Constraints in System and Control Theory</i>
5:00	Bill Pletsch, <i>Investigating Young Group Double Cosets with Computer Algebra: Latest Advances with Some Proofs</i>	Quoc-Nam Tran, <i>A Maple package for fast conversion of Groebner bases</i>	Young Investigator: Virginia Rodrigues, <i>Grobner Basis Structure of Finite Sets of Points.</i>
5:30		Éric Schost, Erwan Le Pennec, <i>Computing foveal wavelets</i>	Young Investigator: Ana Gonzalez-Uriel, <i>Expert System for House Layout Selection</i>
	Activity		

Wednesday July 30, 2003

Start Time	A	B	C
9:00	Redhat Tour 		
9:30			
10:00			
10:30			
11:00	Rida Farouki, <i>Minkowski geometric algebra of complex sets.</i>	Bradford Hovinen, <i>On Montgomery's Block Lanczos Algorithm</i>	Laurent Bernardin, <i>Maple on the Web</i>
11:30	Lihong Zhi, <i>A complete symbolic-numeric linear method for camera pose determination.</i>	Victor Pan, <i>Nearly Optimal Toeplitz/Hankel Computations</i>	Stephen Buswell, <i>MathML, OpenMath and the Evolution of Maths on the Web</i>
12:00	John May, <i>Bounding the Radius of Irreducibility of Multivariate Polynomials.</i>	Shuhong Gao, <i>Random Krylov spaces over finite fields</i>	Olga Caprotti, <i>Mathematical Services Registration and Discovery</i>
12:30	Lunch		
2:00	Mohamed Shalaby, <i>Spline Implicitization of Planar Curves</i>	Wayne Eberly, <i>On the Reliability of Block Lanczos Algorithms</i>	James Davenport, <i>Mathematical Knowledge Management.</i>
2:30	Wen-shin Lee, <i>Symbolic-Numeric Sparse Interpolation of Multivariate Polynomials.</i>	Pascal Giorgi, <i>From BLAS routine to finite field exact linear algebra solution</i>	Mike Dewar, <i>OpenMath and Web Services.</i>
3:00	Bohumir Bastl, <i>Symbolic-numeric method for computing surface self-intersection.</i>	Keith Geddes, <i>Exploiting Fast Hardware Floating Point in High Precision Computation</i>	Mika Seppälä, <i>Advanced Learning Technologies Project.</i>
3:30	Break		
4:00	Ilias S. Kotsireas, <i>Implicit Polynomial Support Optimized for Sparseness.</i>	B. David Saunders, <i>Rank and Smith Form of extremely sparse matrices</i>	Clare M. So and Stephen M. Watt, <i>Conversion Between Content MathML and OpenMath</i>
4:30	Elisabeth Malsch, <i>A symbolic method for defining test functions which satisfy ellipticity, boundary conditions lower order field behaviors exactly.</i>	Robert Lewis, <i>Using the Dixon Resultant on Big Problems</i>	Tom Wickham-Jones, <i>Math on the Web with Mathematica Technology.</i>
5:00	Daniel Lichtblau, <i>Computational algebra visits number theory: Trigonometric polynomials, planar extremal packings, and Groebner bases.</i>	Arne Storjohann, <i>Effective reductions to matrix multiplication</i>	
	Activity		

Thursday July 31, 2003

Start Time	
9:00	Panel Discussion
9:30	
10:00	
10:30	Trip to Wrightsville Beach (see Events Page) 